QP Code: 24900025

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION
NOVEMBER 2024

Multidisciplinary Course-MGIMDCBCH100- SPORTS BIOCHEMISTRY: THE SCIENCE OF EXERCISE AND HUMAN PERFORMANCE

(2024 Admission Onwards)

Scheme for Valuation

Part A
Multiple Choice Questions
Answer all questions. Each question carries 1 mark

1. What is the primary purpose of carbohydrate replenishment after exercise [U][5]
   Answer: Option b-To replenish glycogen stores
2. Find the similarity between cardiac muscles and skeletal muscles [E][3]
   Answer: Option b-Both have striations
3. Which of the following mineral promote the building of strong bones ? [U][2]
   Answer: Option a- Calcium
4. Which of the following is an example of a pharmacological ergogenic aid? [U]][4]
   Answer: Option b- Anabolic Steroids
5. How does sports biochemistry contribute to injury prevention? [K][1]
   Answer: Option b- By analysing biochemical markers for injury risk

(5 x 1= 5)

Part B
Fill in the Blanks
Answer all questions. Each question carries 2 marks

6. The main byproduct of anaerobic glycolysis that leads to muscle fatigue is __ [U][3]
   Answer: Lactic acid
7. One common side effect of beta-alanine supplementation, especially at higher doses, is a tingling sensation on the _____ [K][4]
   Answer: Muscle
8. Mineral which promote the building of strong bones is _____ [K][2]
   Answer: Calcium
9. Cryotherapy involves the use of……to reduce pain and inflammation [K][5]
   Answer: Cold/ice
10. The enzyme _____ is released into the bloodstream when there is damage to skeletal or cardiac muscle. [U][5]
    Answer: Creatine Kinase (5 x 2= 10)

Part C
Short Answer Questions
Answer any 5 questions. Each question carries 3 marks

11. How are cytokines used to assess tissue damage and inflammation in injury? [U][5]
    Answer: Mention about cytokines as a biochemical marker of injury- **1.5 marks**

Discuss the role of cytokines in assessing tissue damage and inflammation in injury - **1.5 marks**

12. How improper hydration affects performance of a sportsman? [U][2]
    Answer: Give the importance of hydration -**1 mark**
    Discuss on the physiological effects of improper hydration- **2 marks**
13. Write a note on important sports drinks and supplements [K][2]
    Answer: Discuss the purpose and benefits of sports drinks-**1.5 marks**
    Purpose and benefits of energy gels, energy supplements and other supplements - **0.5 marks each**
14. Why is carbohydrate consumption important during recovery after endurance sports? [U][3]
    Answer: Mention about the role of nutrition during recovery after endurance sports- **1 mark**
    Role of carbohydrates during recovery after endurance sports-**2 marks**
15. What is the difference between a sprain and a strain? [E][5]
    Answer: Definition, occurrence and symptoms of sprain **1.5 marks**
    Definition, occurrence and symptoms of strain **1.5 marks**
16. What technologies are commonly used in biomechanical assessments to capture movement data? [U][4]
    Answer: Discuss about means for ensuring human performance **1.5 marks**
    Ways to prevent injuries **1.5 marks**
17. How does cellular adaptation affect the number of mitochondria in muscle cells? [U][3]
    Answer: Discuss about mitochondrial biogenesis, increased capillary density, enhanced myoglobin content-**1 mark each**

(5 x 3= 15)

Part D

Short Essay type questions

Answer any one question. Each question carries 5 marks

18. Explain the significance of biomarker monitoring in assessing the progression of injuries? [A][5]
    Answer: Mention about biochemical markers of injury- **1mark**
    Explain how biochemical markers assess the severity and progression of injuries-**4 marks**
19. Appreciate the functions of ergogenic aids in enhancing athletic performance. [Ap][4]
    Answer: Define ergogenic aids- **1 mark**
    list out different types of ergogenic aids- **1 mark**
    functions of ergogenic aids- **3 marks**